

Chapter 35: Predictive Modelling of Personal Loan Acceptance: A Logistic Regression-Based Variable Importance Study

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Abstract

Personal loan acceptance is a critical decision problem for retail banking, requiring a clear understanding of which customer attributes genuinely drive the outcome. This chapter presents a statistical study of personal loan acceptance using the Bank Loan Modelling dataset available on Kaggle, comprising 5,000 customer records described by fourteen demographic and financial attributes. A binary logistic regression model was fitted with loan acceptance as the dependent variable and age, experience, income, family size, average credit card spending, education, mortgage, online banking usage, credit card ownership, securities account status and certificate of deposit (CD) account status as candidate predictors. Multicollinearity among the predictors was assessed using the Variance Inflation Factor (VIF); age and experience showed unacceptably high VIF values and, together with the statistically non-significant mortgage variable, were removed from the model. The reduced logistic regression model was re-estimated, and variable importance analysis was then used to rank the retained predictors by their contribution to the prediction of loan acceptance. Income, education, CD account ownership and family size emerged as the most influential predictors, while online banking usage, securities account ownership and average credit card spending contributed comparatively less. The findings show that combining collinearity diagnostics with logistic regression and variable importance analysis yields a parsimonious and interpretable model that can support evidence-based lending decisions and risk-assessment strategies in retail banking.

Keywords: Logistic Regression, Multicollinearity, Variance Inflation Factor, Variable Importance, Personal Loan

1. Introduction

Loans occupy a central place in modern personal finance, enabling individuals to meet housing, business, educational and consumption needs that could not otherwise be met from current income. For retail banks, the decision to extend a personal loan to a customer is simultaneously a marketing opportunity and a credit-risk problem, and an incorrect decision in either direction is costly: rejecting a creditworthy customer forgoes revenue, while approving a high-risk customer increases the likelihood of default. Statistical modelling of the loan-acceptance decision therefore remains an important area of applied research in banking and finance.

Binary outcomes such as loan acceptance are most naturally modelled using logistic regression, a technique introduced by Cox (1958) and now a standard tool in applied statistics for relating a dichotomous response to a set of predictors (Hosmer, Lemeshow, & Sturdivant, 2013; Menard, 2002). Unlike ordinary least squares regression, logistic regression models the log-odds of the outcome as a linear function of the predictors, which keeps predicted probabilities bounded between 0 and 1 and allows coefficients to be interpreted in terms of odds ratios. This property has made logistic regression a

preferred baseline model in the credit-scoring literature, even as more flexible machine-learning classifiers such as random forests (Breiman, 2001) and gradient-boosted trees have become popular alternatives (James, Witten, Hastie, & Tibshirani, 2013).

A growing body of empirical work has applied logistic regression, alone or alongside machine-learning classifiers, to loan and credit-risk datasets. Kwofie, Owusu-Ansah, and Boadi (2015) used binary logistic regression to model the probability of loan default and identified income and repayment history as dominant predictors, while Alagic et al. (2024) compared logistic regression with ensemble classifiers on loan-approval data and reported that, despite the appeal of more complex algorithms, logistic regression remained competitive and considerably easier to interpret for lending decisions. These findings motivate the continued use of logistic regression as an interpretable benchmark model, particularly when a bank must justify individual lending decisions to regulators or customers.

A recurring difficulty in fitting logistic regression models to financial and demographic data is multicollinearity among predictors, which inflates the standard errors of the regression coefficients and destabilises their estimated effects (Gujarati & Porter, 2009). The Variance Inflation Factor (VIF) is the most widely used diagnostic for this problem, and although the conventional threshold of 10 is a rule of thumb rather than a formally derived cut-off, it remains a practical and widely accepted screening criterion in applied regression work (O'Brien, 2007). Once collinear or statistically non-significant predictors have been identified, variable importance analysis provides a complementary, coefficient-independent way of ranking the remaining predictors by their overall contribution to the model, which is particularly useful for communicating results to non-technical stakeholders.

Building on this literature, the present chapter uses the publicly available Bank Loan Modelling dataset hosted on Kaggle (Kaggle, n.d.) to develop an interpretable logistic regression model of personal loan acceptance. The specific objectives are (i) to diagnose multicollinearity among the candidate predictors using VIF and to arrive at a reduced, stable predictor set; (ii) to re-estimate the logistic regression model on this reduced set and assess the statistical significance of each retained predictor; and (iii) to rank the retained predictors using variable importance analysis in order to identify the factors that most strongly influence the likelihood that a customer accepts a personal loan offer.

2. Materials and Methods

Data Source

The dataset used in this study is the Bank Loan Modelling dataset available on Kaggle (<https://www.kaggle.com/datasets/itsmesunil/bank-loan-modelling/data>), which contains 5,000 customer records described by 14 variables, including demographic attributes (age, experience, family size, education), financial attributes (income, average credit card spending, mortgage value) and indicators of banking-product usage (online banking, credit card ownership, securities account and certificate of deposit account). The dependent variable, Personal Loan, records whether a customer accepted a personal loan offer (1) or not (0). Customer ID and ZIP code were excluded from the model as they carry no explanatory information about creditworthiness.

Model Specification

The relationship between the log-odds of loan acceptance and the predictor variables was modelled using binary logistic regression:

$$P(Y = 1|X) = \frac{1}{1+e^{-(\beta_0+\beta_1X_1+\beta_2X_2+\dots+\beta_nX_n)}} \quad (1)$$

where $Y = 1$ denotes acceptance of the personal loan, $X_1, X_2, \dots, X_n$ are the predictor variables, and $\beta_0, \beta_1, \dots, \beta_n$ are coefficients estimated by maximum likelihood.

Multicollinearity Diagnostics

Multicollinearity among predictors was assessed using the Variance Inflation Factor:

$$VIF_i = \frac{1}{(1-R_i^2)} \quad (2)$$

where R_i^2 is the coefficient of determination obtained by regressing the i -th predictor on all remaining predictors. A VIF exceeding 10 was treated as evidence of problematic multicollinearity (O'Brien, 2007), and predictors flagged in this way, together with any predictor lacking statistical significance ($p > 0.05$), were dropped before the final model was re-estimated.

Variable Importance Analysis

The relative contribution of each retained predictor to the reduced logistic regression model was assessed using variable importance analysis, which scales each predictor's effect to a common metric and ranks predictors independently of the raw coefficient magnitudes. All computations were performed in R using standard regression and model-evaluation routines.

3. Results

Multicollinearity Assessment

Table 1 reports the VIF for each of the eleven candidate predictors of personal loan acceptance.

Table 1: Variance Inflation Factor (VIF) for candidate predictors

Variable	VIF
Age	91.71
Experience	91.57
Income	2.43
Family	1.32
CCAvg	1.44
Education	1.91
CD Account	2.07
Mortgage	1.04
Online	1.13
Credit Card	1.40
Securities Account	1.37

Age and Experience recorded VIF values close to 92, far exceeding the conventional threshold of 10, indicating severe multicollinearity between the two variables. All remaining predictors had VIF values below 2.5, indicating negligible collinearity. Age, Experience and the statistically non-significant Mortgage variable ($p = 0.392$ in the full model) were therefore excluded, and the logistic regression

model was re-estimated on the remaining eight predictors: Income, Family, CCAvg, Education, CD Account, Online, Credit Card and Securities Account.

Reduced Logistic Regression Model

Table 2: Estimated coefficients of the reduced logistic regression model

Variable	Estimate (β)	Std. Error	p-value
Intercept	-13.224	0.562	< 2e-16
Income	0.0547	0.0026	< 2e-16
Family	0.6904	0.0742	< 2e-16
CCAvg	0.1137	0.0393	0.0038
Education	1.7041	0.1124	< 2e-16
CD Account	3.8533	0.3234	< 2e-16
Online	-0.6675	0.1567	2.05e-05
Credit Card	-1.1237	0.2050	4.22e-08
Securities Account	-0.9346	0.2848	0.0010

All eight retained predictors were statistically significant ($p < 0.01$). Income, Family, CCAvg, Education and CD Account carried positive coefficients, indicating that higher income and education levels, larger family size, higher average credit card spending and CD-account ownership were all associated with a higher likelihood of accepting the personal loan offer. Online banking usage, credit card ownership and securities-account ownership carried negative coefficients, indicating a lower likelihood of loan acceptance among customers who already used these other banking products.

Variable Importance

Table 3: Relative variable importance for personal loan acceptance

Variable	Importance score
Income	21.13
Education	15.16
CD Account	11.91
Family	9.30
Credit Card	5.48
Online	4.26
Securities Account	3.28
CCAvg	2.90

Income was the single most influential predictor of personal loan acceptance, followed by Education, CD Account and Family, together accounting for the majority of the model's explanatory contribution. Credit Card, Online and Securities Account contributed moderately, while CCAvg was the least influential of the retained predictors.

4. Discussions

The strong collinearity between Age and Experience is unsurprising, since years of professional experience are mechanically bound to increase with age; retaining both variables in a single model would have produced unstable and difficult-to-interpret coefficients without adding genuinely new information. Removing the pair, together with the non-significant Mortgage variable, produced a more parsimonious model without any meaningful loss of explanatory content, consistent with the general guidance that variables flagged by a high VIF together with weak significance are reasonable candidates for exclusion (O'Brien, 2007; Gujarati & Porter, 2009).

The dominance of Income, Education, CD Account and Family among the retained predictors is broadly consistent with prior applied work linking loan-related decisions to income and financial-literacy proxies such as education (Kwofie, Owusu-Ansah, & Boadi, 2015; Alagic et al., 2024). The negative association between existing product ownership (Online banking, Credit Card, Securities Account) and personal loan acceptance is a less intuitive but practically important finding: customers who are already engaged with other banking products may either have less unmet demand for additional credit or may represent a different underlying customer segment than those who accept a personal loan offer. This pattern illustrates the value of variable importance analysis as a complement to raw coefficients, since importance rankings do not depend on the sign of the effect and instead highlight the predictors that most strongly discriminate between acceptors and non-acceptors.

The chapter's findings should be read in light of some limitations. The dataset is cross-sectional and drawn from a single institution's historical campaign, so the estimated coefficients may not generalise to other lending contexts or time periods. The analysis also relies on a purely statistical model; comparing it against ensemble machine-learning classifiers, as in Alagic et al. (2024) and James, Witten, Hastie, and Tibshirani (2013), would help establish whether the gain in predictive accuracy from more flexible models is large enough to offset their reduced interpretability in a regulated lending environment.

5. Conclusions

This chapter demonstrated a complete statistical workflow for modelling personal loan acceptance: diagnosing multicollinearity with the Variance Inflation Factor, removing collinear and non-significant predictors, re-estimating a logistic regression model on the reduced predictor set, and ranking the retained predictors using variable importance analysis. Income, Education, CD Account ownership and Family size emerged as the strongest drivers of personal loan acceptance, while Age, Experience and Mortgage added little beyond redundant or non-significant information. The resulting model is both statistically sound and readily interpretable, making it a suitable basis for lending policy, targeted marketing of loan products, and risk-assessment practice in retail banking. Future work could extend this analysis by benchmarking the logistic regression model against ensemble machine-learning classifiers (Breiman, 2001) and by validating the model on data drawn from other institutions or time periods.

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